

Tien-Ning(Sky) Lee

Email: tienninglee@gmail.com | TEL: +1 858-214-8732 | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of California, San Diego (UCSD)

M.S. in Electrical and Computer Engineering (Machine Learning & Data Science Track)

Sep. 2025 - Expected Jun. 2027

California, United States

National Yang Ming Chiao Tung University (NYCU)

B.S. in Information Management and Finance (Program of Information Management)

Sep. 2020 - Jun. 2024

Hsinchu, Taiwan

EXPERIENCE

Classmethod

Jul. 2026 - Sep. 2026

Physical AI Engineer Intern

Tokyo, Japan

- Developed Physical AI applications for **Unitree G1 and Go2** robots, including a visual inspection agent that converts live camera input into object detections, risk assessments, and operator alerts.
- Integrated **Unitree SDK2**, **CycloneDDS**, and persistent GPU inference services for continuous camera ingestion, structured outputs, deterministic geometry policies, and automated inspection reports.
- Profiled the end-to-end vision pipeline and optimized image processing and camera capture, **reducing overlay latency by 92%** and improving Go2 image updates from 3–4 seconds to approximately 1 second.
- Evaluated vision-language models and built a **Grounding DINO** monitoring pipeline with ROI-based risk analysis, real-time overlays, inference telemetry, process controls, and audio warnings.

Artificial Intelligence Lab, National Taiwan University

Jun. 2024 - Dec. 2024

Project Research Assistant | Advisor: Prof. Tsung-Nan Lin (IEEE Fellow)

Taipei, Taiwan

- Co-developed **trustworthy AI recommendation systems** with PhD students for **BankTaiwan Life Insurance**, focusing on minimizing default risk through **anti-recommendation algorithms**.
- Processed large-scale real-world insurance data and **engineered robust features** to enhance model generalization.
- Implemented and evaluated collaborative filtering and **anomaly detection** models using **PyTorch**, optimizing for both ranking accuracy and risk mitigation.
- Collaborated with stakeholders to **officially deploy** the model after successful enterprise testing.

National Institute of Information and Communications Technology (NICT)

Jul. 2023 - Sep. 2023

AI Research Intern

Tokyo, Japan

- Conducted research to enhance the resilience of **model-based network intrusion detection systems (NIDS)** for **IoT devices** by investigating advanced **adversarial attacks** to develop robust defense strategies.
- Engineered and trained diverse NIDS models—including **Logistic Regression, kNN, Random Forest, Autoencoder, 1D-CNN, and 2D-CNN**—using **scikit-learn** and **TensorFlow** on **real-world TON_IoT datasets**.
- Developed a **SHAP**-guided feature selection algorithm that **reduced input features by 65%** while maintaining model performance and interpretability.
- Investigated adversarial training using **SHAP**-informed attacks, providing key insights that supported NICT researchers' ongoing studies on adversarial training and defense strategies.

PROJECTS

Amazon-SASRec-Engine | PyTorch, FAISS, FastAPI, Streamlit | Full-Stack Deep Recommender System

- Developed a **full-stack sequential recommendation system** using the **SASRec** (Transformer) model to predict user purchase intent based on interaction history.
- Implemented **FAISS** to achieve **sub-millisecond** inference latency for **real-time** recommendations.
- Designed a decoupled architecture with **Streamlit** (frontend) and **FastAPI** (backend), featuring an oversampling strategy to filter duplicates and ensure consistent top-k results.
- Integrated a real-time web scraper with **BeautifulSoup** as a fallback mechanism to dynamically fetch missing product metadata from Amazon, resolving cold-start display issues.

SKILLS

Languages: Mandarin Chinese (native), English (fluent), Japanese (business level, JLPT N2)

Programming Languages: Python, C++, JavaScript, Java, R, SQL

Libraries/Frameworks/Tools: scikit-learn, TensorFlow, PyTorch, React, FastAPI, Alembic, Git, AWS, Render, Docker, ROS2, pytest